

Universidade Federal de Santa Catarina
UFSC - Campus Blumenau

INTRODUÇÃO À ROBÓTICA INDUSTRIAL

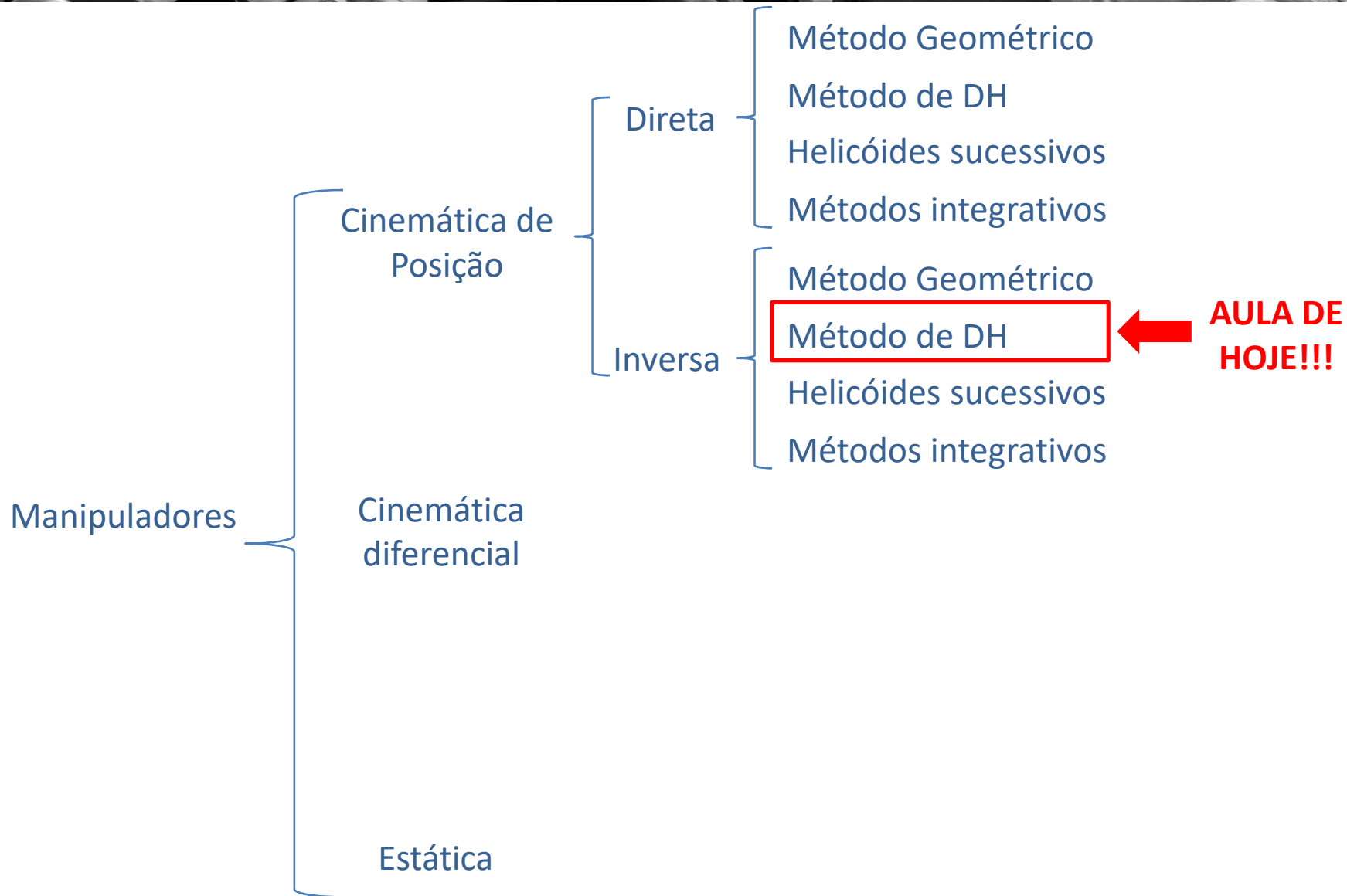


Prof. Leonardo Mejia Rincon
leonardo.mejia.rincon@ufsc.br



MOTIVAÇÃO

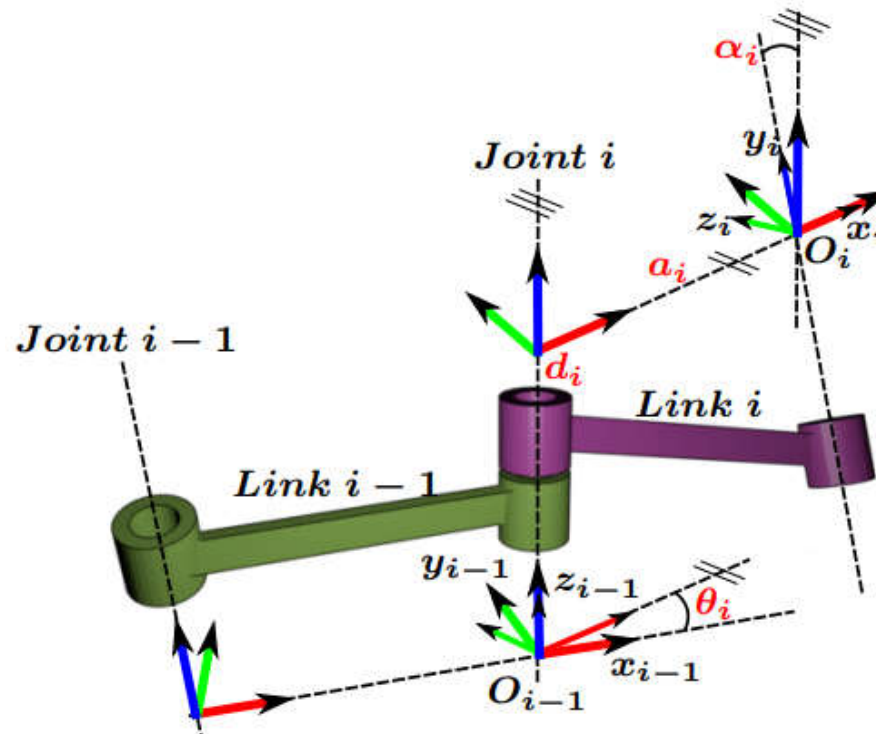




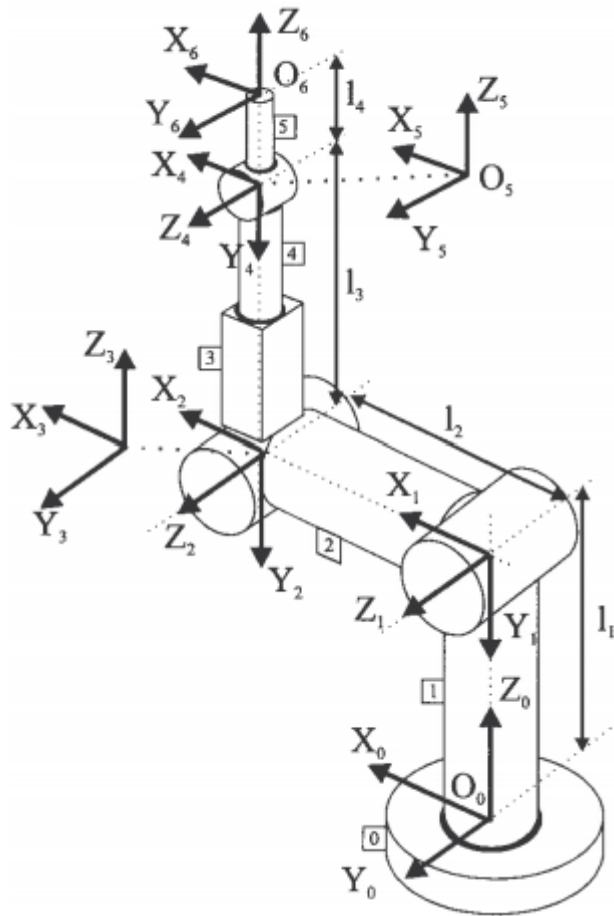
Método de Denavit and Hartenberg (DH):

Os quatro parâmetros de transformação seguintes são conhecidos como parâmetros DH:

- d : Distância ao longo do “z” anterior até a normal comum;
- θ : Ângulo em torno do “z” anterior, do “x” anterior até o “x”;
- a : comprimento da normal comum;
- α : ângulo em torno da normal comum, do “z” anterior até o “z”



Método de Denavit and Hartenberg (DH):



$${}^{i-1}T_i = \begin{bmatrix} c\theta_i & -s\theta_i \cdot c\alpha_i & s\theta_i \cdot s\alpha_i & a_i \cdot c\alpha_i \\ s\theta_i & c\theta_i \cdot c\alpha_i & -c\theta_i \cdot s\alpha_i & a_i \cdot s\alpha_i \\ 0 & s\alpha_i & c\alpha_i & d \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



MATRIZ “NOAP”

Normal-Orientation-Approach-Position

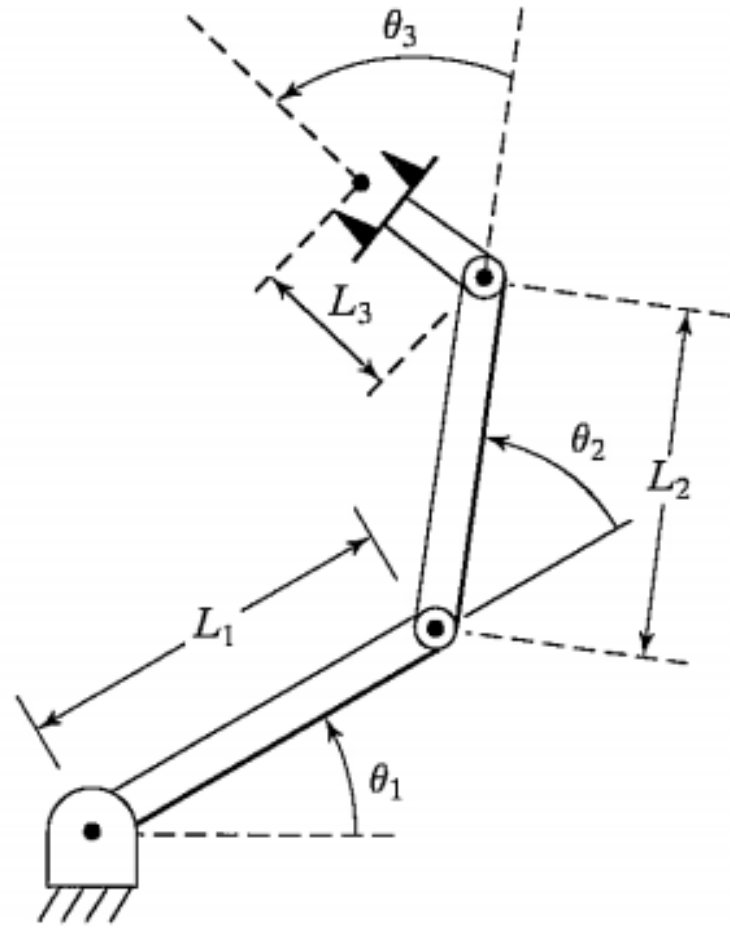
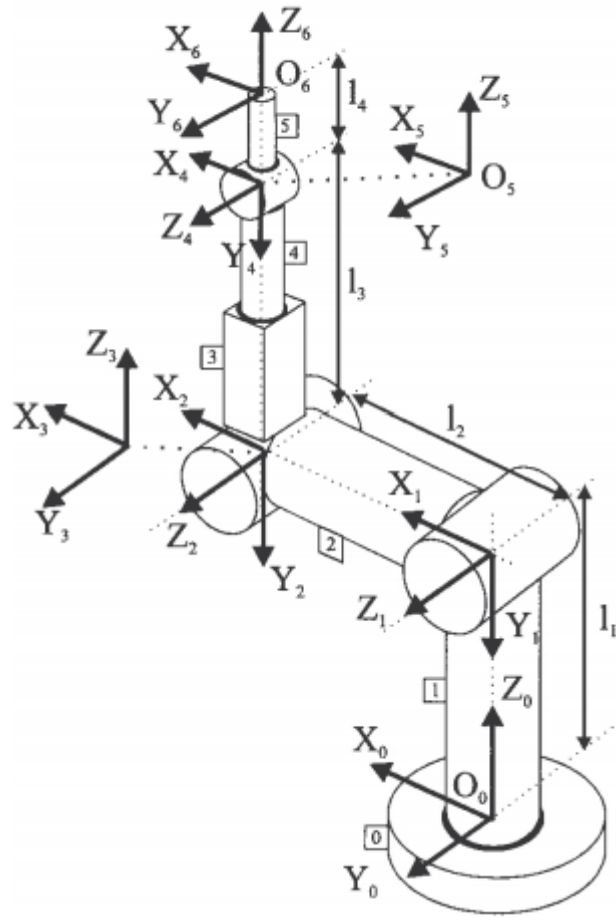




Método de Denavit and Hartenberg (DH):



Método de Denavit and Hartenberg (DH):





INVERSA DE UMA MATRIZ DE TRANSFORMAÇÃO HOMOGÊNEA





Método de Denavit and Hartenberg (DH):

$$\begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} n_x & n_y & n_z & -\mathbf{n}^T \mathbf{p} \\ o_x & o_y & o_z & -\mathbf{o}^T \mathbf{p} \\ a_x & a_y & a_z & -\mathbf{a}^T \mathbf{p} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

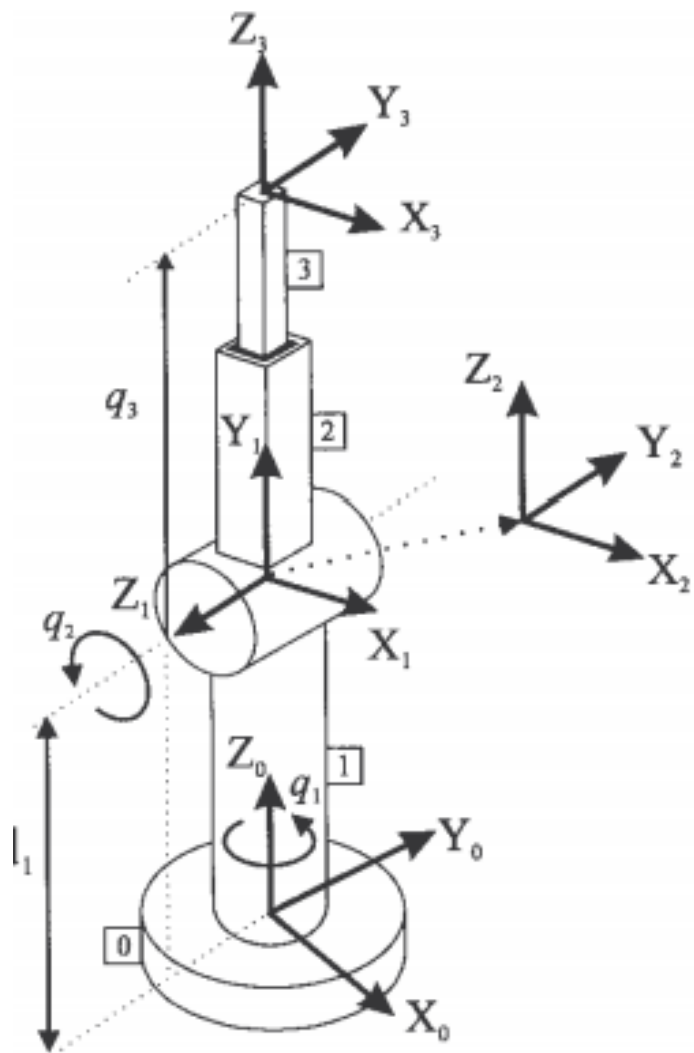
$${}^0\mathbf{A}_1^{-1} = \begin{bmatrix} C_1 & 0 & S_1 & 0 \\ S_1 & 0 & -C_1 & 0 \\ 0 & 1 & 0 & l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} C_1 & S_1 & 0 & 0 \\ 0 & 0 & 1 & -l_1 \\ S_1 & -C_1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$




EXEMPLOS DE APLICAÇÃO DO MÉTODO DE DENAVIT AND HARTENBERG



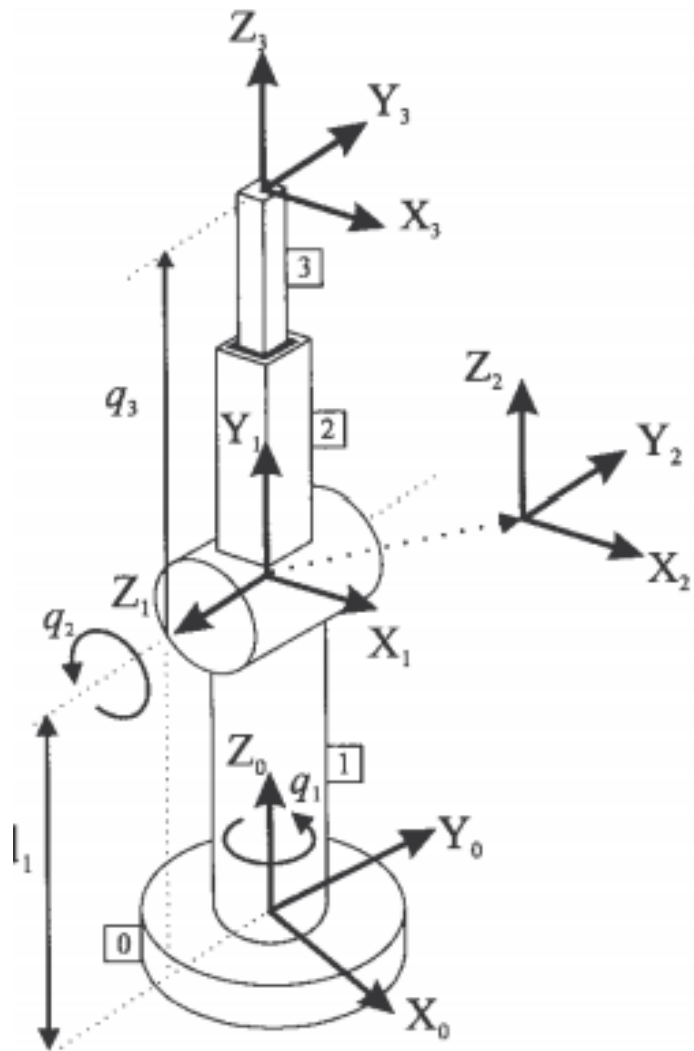
Método de Denavit and Hartenberg (DH):



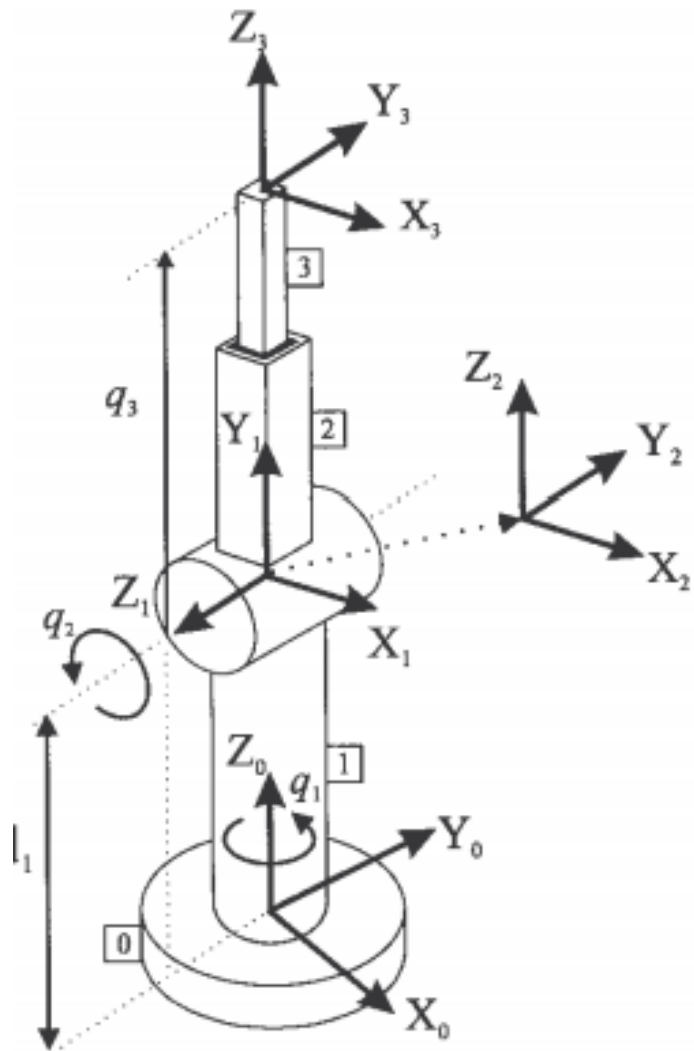
i	α_i	a_i	d_i	θ_i

$${}^{i-1}T_i = \begin{bmatrix} c\theta_i & -s\theta_i \cdot c\alpha_i & s\theta_i \cdot s\alpha_i & a_i \cdot c\alpha_i \\ s\theta_i & c\theta_i \cdot c\alpha_i & -c\theta_i \cdot s\alpha_i & a_i \cdot s\alpha_i \\ 0 & s\alpha_i & c\alpha_i & d \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

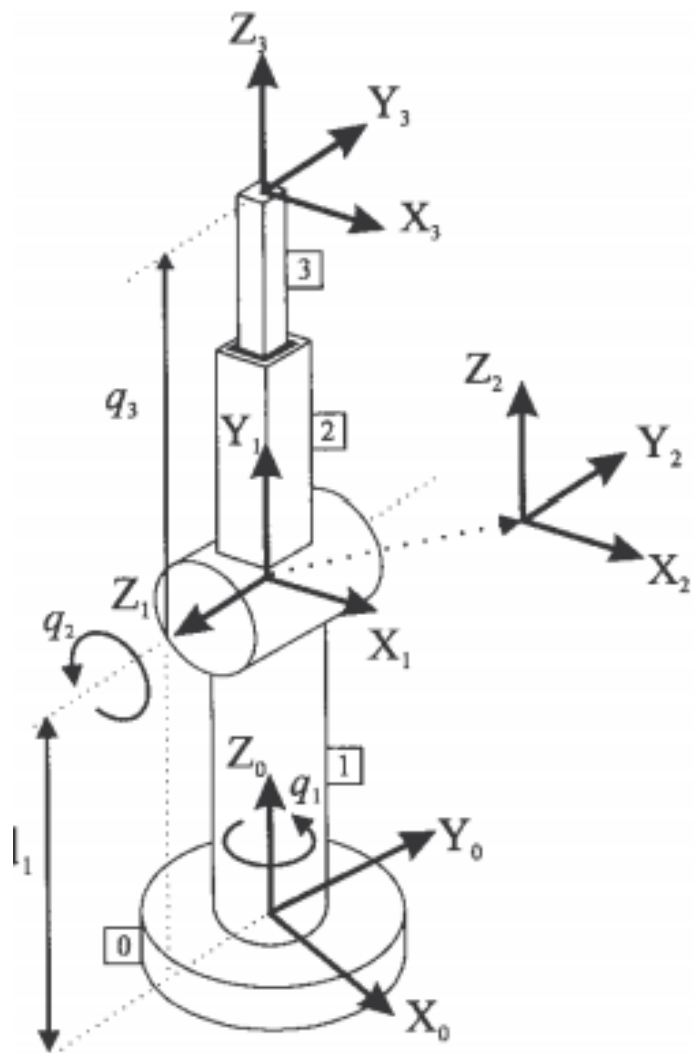
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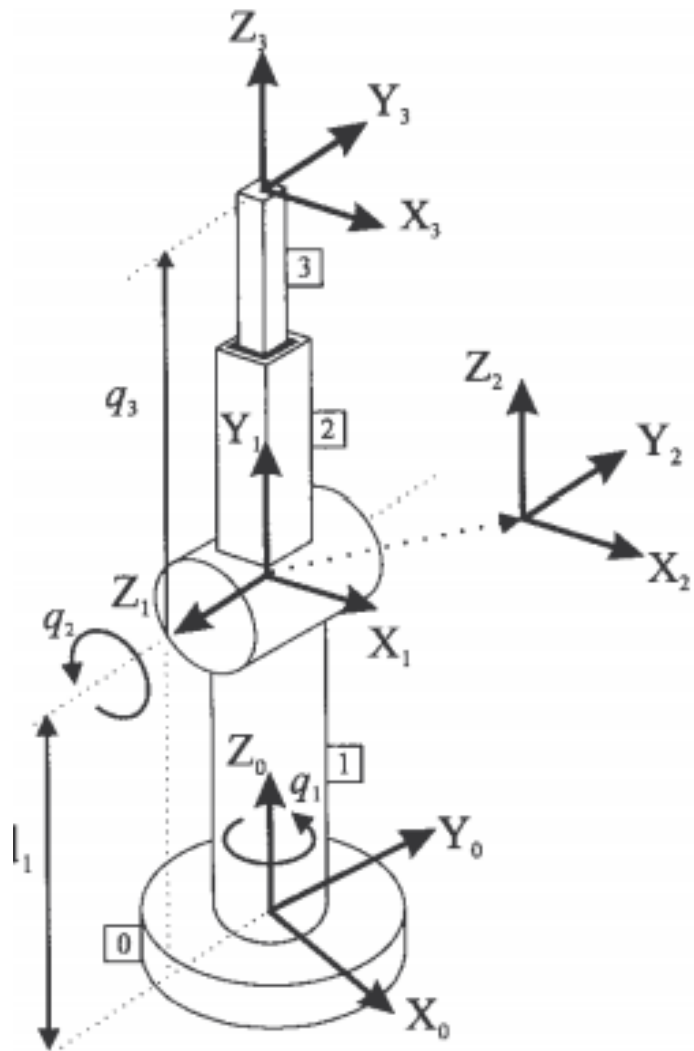
Método de Denavit and Hartenberg (DH):



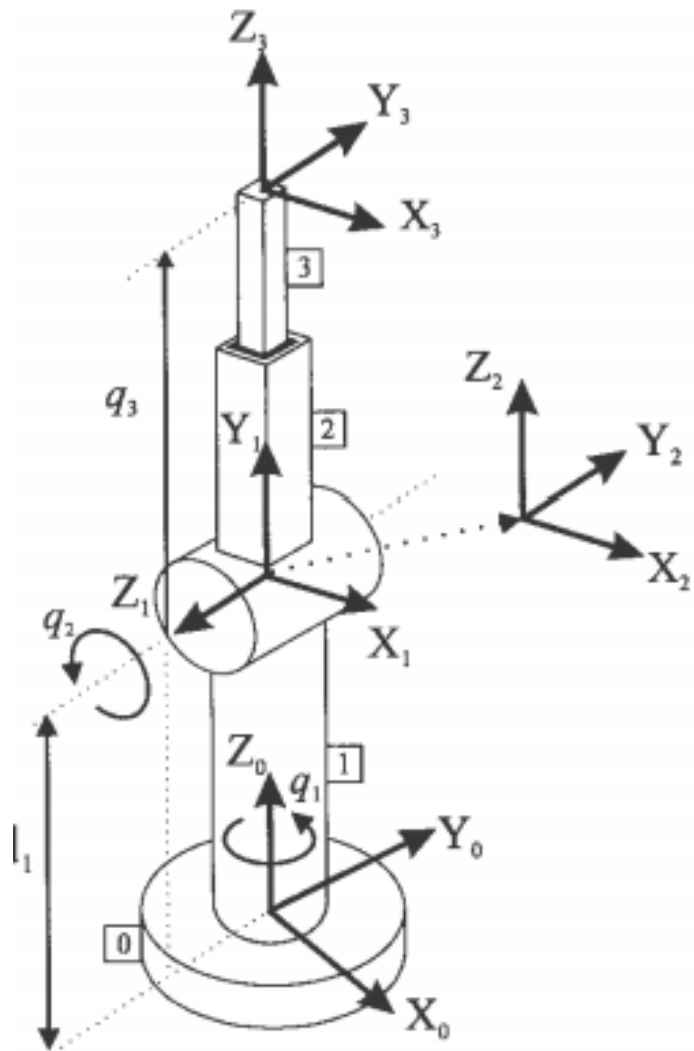
$$({}^0\mathbf{A}_1)^{-1} {}^0\mathbf{T}_3 = {}^1\mathbf{A}_2 {}^2\mathbf{A}_3 = \begin{bmatrix} C_1 & S_1 & 0 & 0 \\ 0 & 0 & 1 & -l_1 \\ S_1 & -C_1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} =$$

$$= \begin{bmatrix} C_2 & 0 & -S_2 & 0 \\ S_2 & 0 & C_2 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & q_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} C_2 & 0 & -S_2 & -S_2 q_3 \\ S_2 & 0 & C_2 & C_2 q_3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

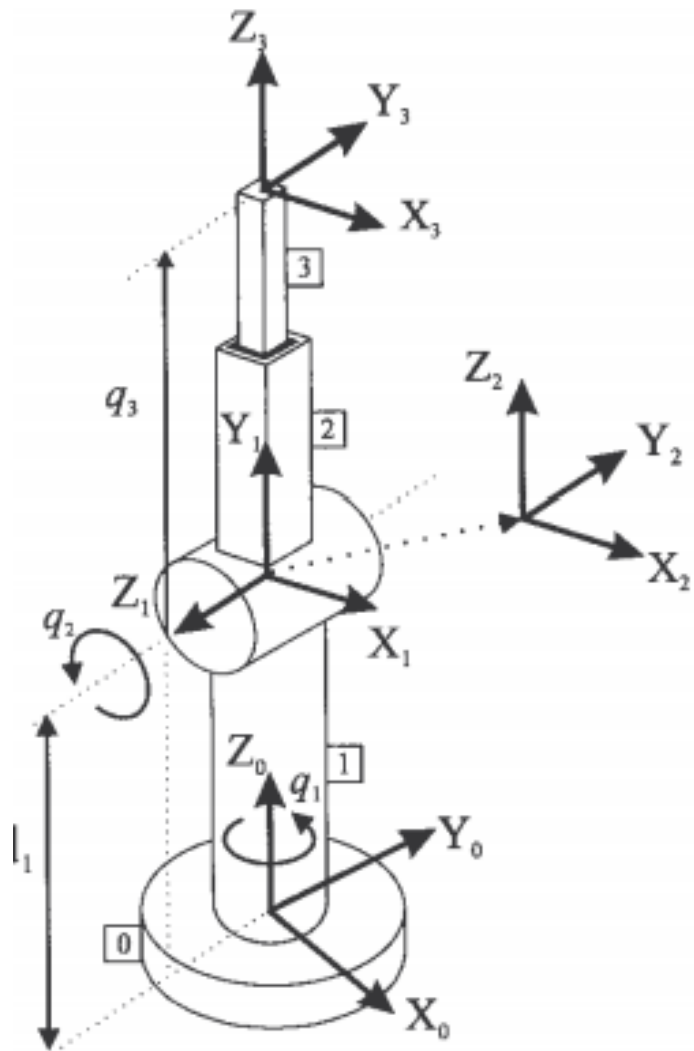
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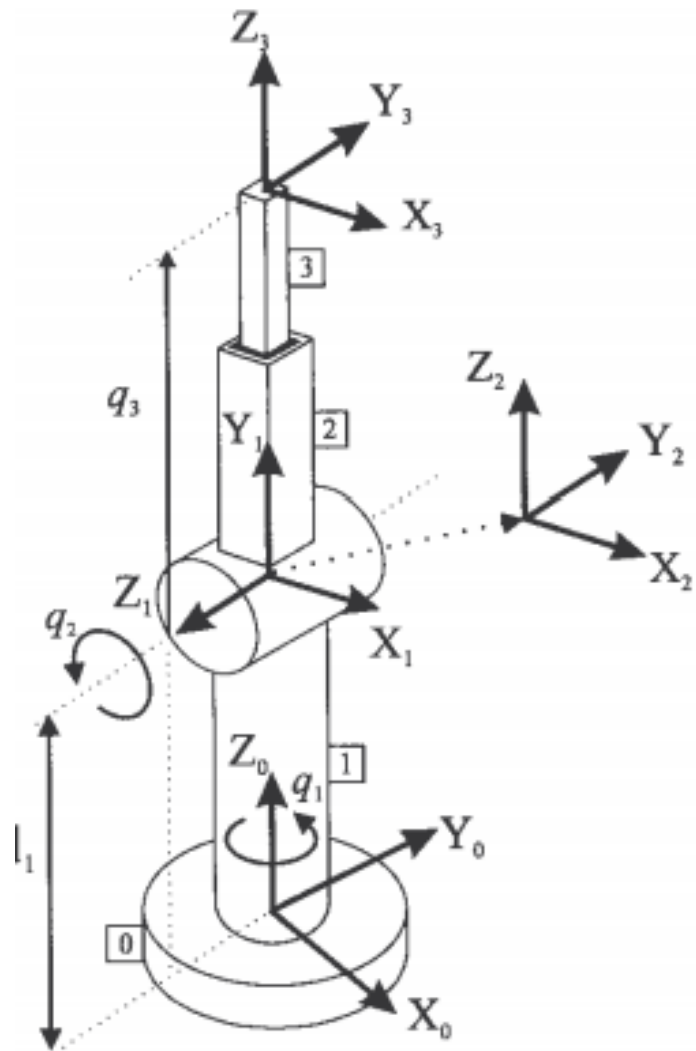
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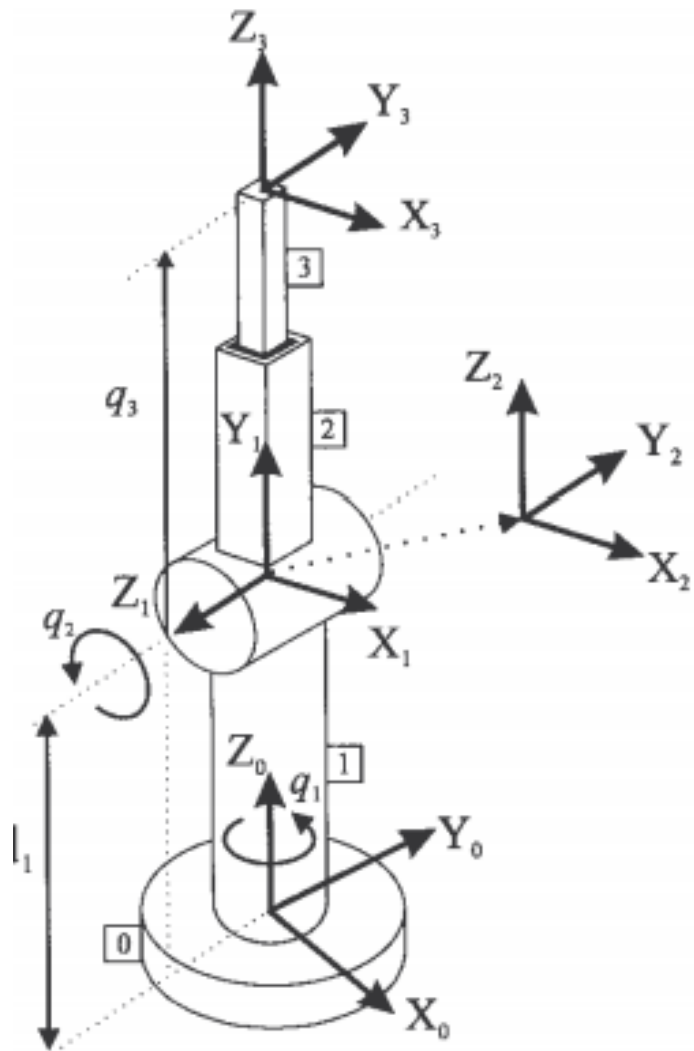


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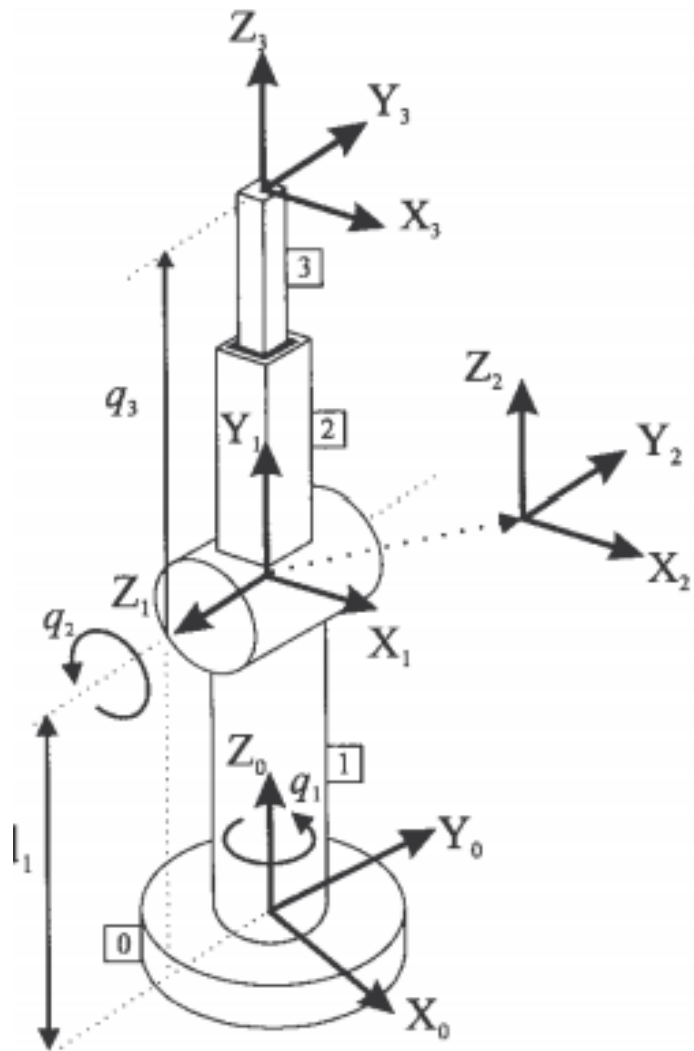


$$\begin{aligned}
 &({}^1A_2)^{-1}({}^0A_1)^{-1}T^2A_3 = \\
 &= \begin{bmatrix} C_2 & S_2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -S_2 & C_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} C_1 & S_1 & 0 & 0 \\ 0 & 0 & 1 & -l_1 \\ S_1 & -C_1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & q_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} C_2C_1 & C_2S_1 & S_2 & -l_1S_2 \\ -S_1 & C_1 & 0 & 0 \\ -S_2C_1 & -S_2S_1 & C_2 & -C_2l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & q_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

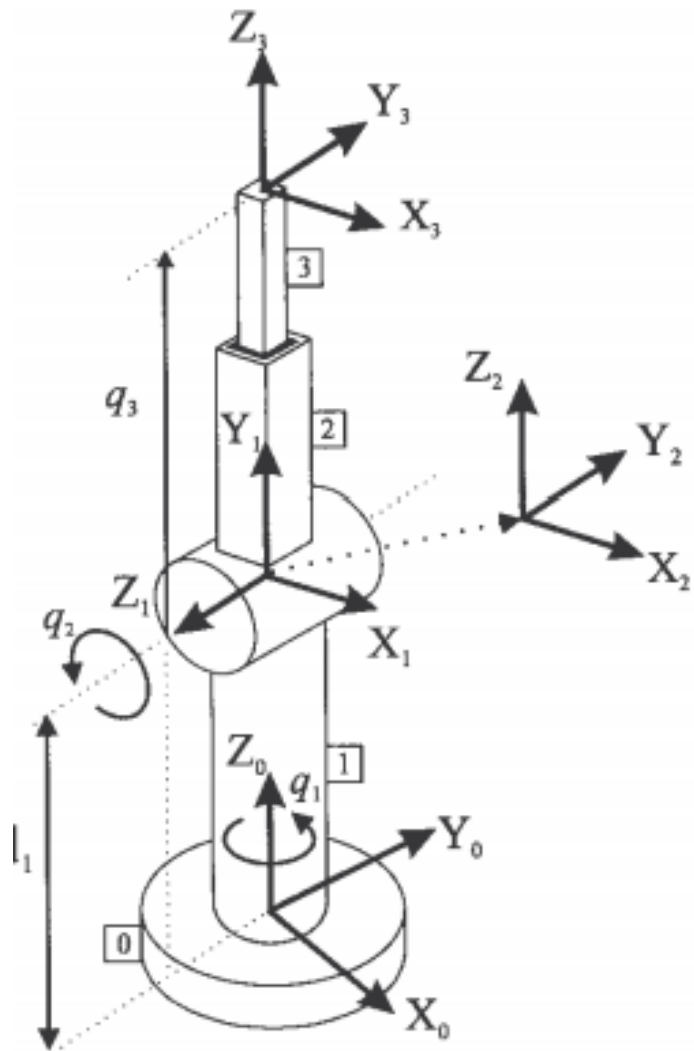
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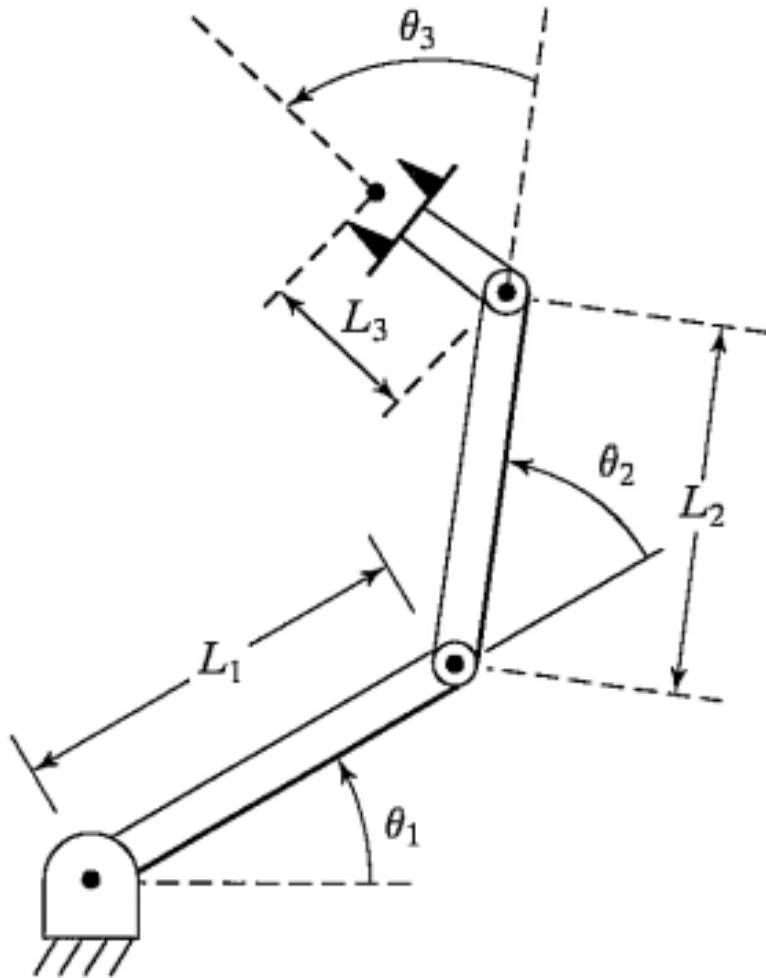
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Fim da Aula



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